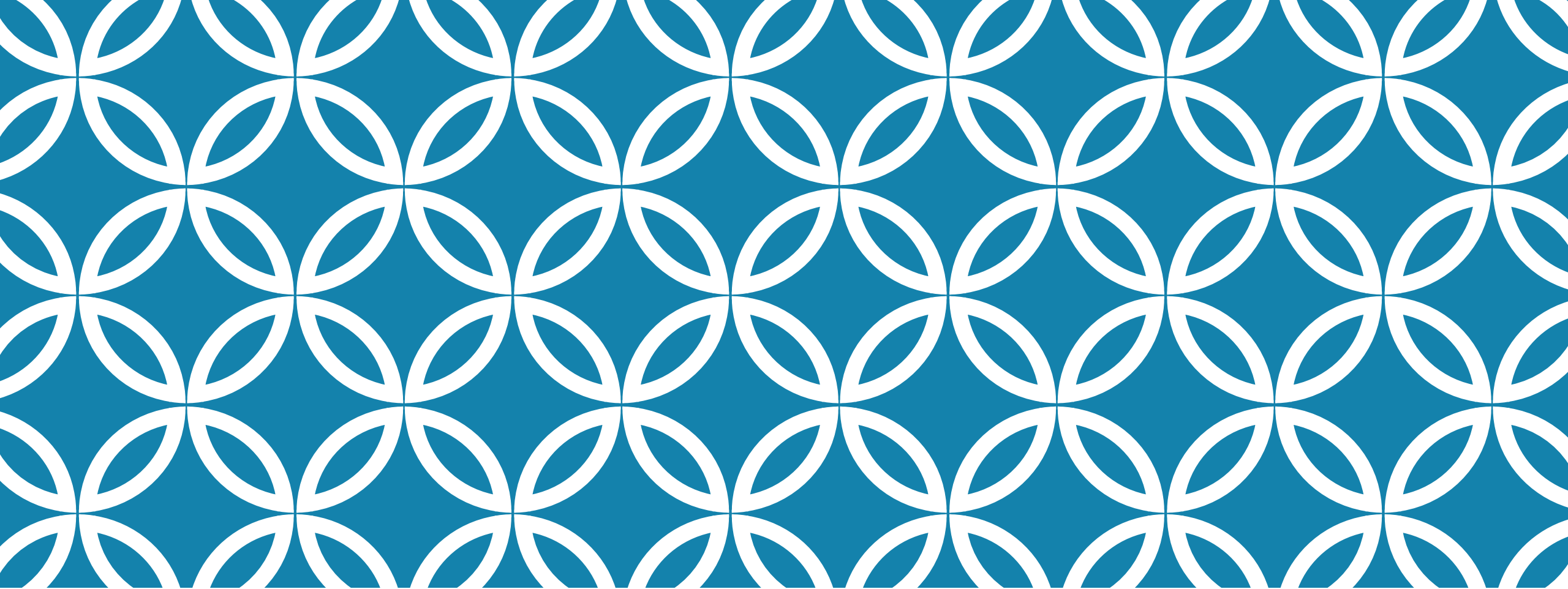




CSE455 - SOFTWARE TESTING

Chapter 4

INTRODUCTION

Analysis of work products (documentation and code) contributes measurably to increased product quality

Static testing can be performed in a tool-based environment or manually and it is often neglected

It can be close examination by one or more persons or can be performed using appropriate analysis tools

In safety-critical industries like aviation and medicine, static testing is an extremely important factor in ensuring the required product quality

STATIC TESTING IS ALL ABOUT PREVENTION

Deviation from plan are identified before they can have a negative effect

All relevant work products are quality assured before anyone does any further work on them

WHAT CAN WE ANALYZE AND TEST?

Quality of each individual document contributes to the overall quality of the results

Specification errors need to be found before they are converted into code

Static testing reveals include inconsistencies, ambiguities, contradictions, gaps, inaccuracies, redundancies.

Source code can be checked for consistency with the project's programming guidelines

It is recommended to also test and verify documents created during testing like test plan, test cases, test procedures and scripts, acceptance criteria

Hence, static testing can improve the overall development process

STATIC TESTING TECHNIQUES

Human mental and analytical skills can be used to analyze and evaluate complex situations

For the success of analysis process, people involved must understand the documents they are reading and comprehend the statements and definitions they contain

Review is the most important static testing technique

Review can be formal (adhere to pre-defined process and participants document a planned set of results) or informal (no clear definition of intended results or what is logged)

Type of review process you choose depends on development model, its maturity, complexity of documents to be tested, and regulatory requirements

REVIEW PROCESS ACTIVITIES



PLANNING

Project management decides which type of review is required for a specific document, its associated estimated time

Project manager selects skilled participants and assign them their roles

Formal review requires predefined entry and exit criteria. If plan includes entry criteria, you need to check (checklist) that these have been fulfilled before review

Different viewpoints of participants increased effectiveness of review process

Often entire document is not required to be reviewed but only the part which contains high risk defects

When and where review meeting will be conducted is also decided

The quality characteristics under investigation are defined

INITIATE REVIEW

All the resources are allocated; participants are provided with all necessary physical and electronic materials

Along work products that are to be reviewed, review participants require other material known as baseline document(s); These include any documents that help to decide whether what they are looking at is a deviation

INDIVIDUAL REVIEW PREPARATION

A review can only be successful if all participants are well prepared

Any potential defects, recommendations, questions or comments are noted

ISSUE COMMUNICATION AND ANALYSIS

Findings are collated and discussed

It can take place at a review meeting or a company-internal online forum

Who is responsible for the correction of which defect, how to monitor correction process and determine whether a follow-up review of fixed document is required

Each quality characteristic is evaluated and the results of the analysis are documented

Review team has to provide a recommendation regarding acceptance of the review object:

- Accepted without changes
- Revision necessary due to extensive changes
- Not accepted (rejected)

FIXING AND REPORTING

Defects found are communicated to responsible person

Correction of defects or inconsistencies the review has revealed

Author rectifies (corrects) the revealed defects

Formal review requires more work; you need to record the current status of a defect or its report, check whether exit criteria met or not

Results of a review process vary considerably depending upon the type of review and degree of formality involved

INDIVIDUAL REVIEW TECHNIQUES

The basic review process requires each participant to prepare for the team review

There are different individual review techniques, their effectiveness depends on the type of review you are conducting:

- Ad hoc
- Checklist based
- Using scenarios and dry runs
- Role and viewpoint based

AD HOC – NO RULES

No predefined rules regarding individual preparation

Reviewer reads the work products sequentially and notes any issues, findings or defects

If a review object is reviewed by more than one person, some issues are sure to be flagged multiple times

CHECKLIST-BASED

Contains a list of questions based on potential defects that have been discovered in the course of earlier projects

The questions can be based on formal guidelines or standards

Must be updated regularly according to newly organized issues and remove questions related to outdated or resolved issues

Should be kept short and only include key questions

It focuses on specific class of questions ignoring other possibilities

Always try to keep an open mind when using this technique

USING SCENARIOS AND DRY RUNS

Scenario-based review is based on predefined, structured guidelines that dictate how to run through the review process

Availability of specific use cases simplifies this process as all you have to do is perform dry runs of all use cases; also check whether each use case is implemented or not in code review

In the case of code review, scenarios can be based on classes of defects like one reviewer focusing on error handling while other concentrates on checking boundaries

Exclusive use of scenarios makes it difficult to identify other types of defects and should be avoided if possible

ROLE AND VIEWPOINT-BASED

The reviewer is tasked with checking the review object from a specialist's point of view; utilize skills each role provides

If end-users or the system's operator cannot take part in review sessions, these roles have to be played by others

If the system is to be used within the organization, a reviewer can play the role of system or user administrator

In role-based review, each reviewer pretends to be someone with a specific job and checks the document from that role's point of view

In viewpoint-based (perspective-based) review, each person looks at the document from their own stakeholder's point of view; understand the needs and concerns of different groups and how they affect decision like priority, effort or risk

ROLES AND RESPONSIBILITIES WITHIN THE REVIEW PROCESS

Not every role has to be filled by one person, and there are overlaps between roles that justify having one person fill multiple roles (for example, moderator and review leader)

Which role can be combined depends on the nature of the project and quality criteria you need to achieve

The individual activities associated with individual roles also vary according to the type of review you are conducting:

- Management
- Review Leader
- Facilitator/Moderator
- Author
- Reviewer
- Scribe

MANAGEMENT

Responsible for planning and allocating the required resources (people, time, money) and keep an eye on cost effectiveness

Selects which work products and supporting materials are to be included in review process

REVIEW LEADER

Describes who participates in a review, as well as when and where it takes place

Holds the overall responsibility and needs to make sure planning, preparation, execution, revision, and any follow-up work all contribute to achieving the review objectives

FACILITATOR/MODERATOR

Ensures the smooth running of review process

Must have good degree of assertiveness and ability to detect undertones within the conversation, good at bringing opposing viewpoints and keeping discussion brief without hurting anyone's feelings

Introduces the participants and their roles, and also provide brief summary of the subject due for inspection

Asks all the reviewers if they are sufficiently well prepared (for instance, by making sure that all checklist questions have been answered)

AUTHOR

Writer of the review object

Ensures that entry and exit criteria are fulfilled

Responsible for rectifying any defects found during the review

Must never take criticism personally

Hence, review process serves solely to improve the quality of the review object

REVIEWER

Participates in the review meeting following preparation

Identify and describe problematic places in the review object

Having different perspectives (tester, developer, end-user, system operator) aids the effectiveness of a review

Individual reviewer concentrates on specific aspect of the review increases the efficiency of the review process like one reviewer concentrates on adherence to a particular standard while another checks the program syntax

Must be able to clearly differentiate between parts of the review object that pass the review and those that require improvement

SCRIBE (RECORDER)

Documents the unresolved issues, open questions, any other related tasks generated by the review

Documentation has to be brief and to include undistorted summaries of all the discussion that took place

For practical reasons, author is often also the scribe because he knows exactly what needs documenting in order to perform changes requested by the reviewer

As dedicated tools become more common, the job of the review scribe is slowly becoming obsolete

TYPES OF REVIEW

Two types of review can be identified when looking at review objects:

- Review of documents that are created as work products *during development process*
- Review that *analyze the project* or development process itself (management review)

Main objective is to identify defects by investigating documents

Can be informal review, a walkthrough, a technical review or an inspection

All of these can be conducted as a peer review in which participation of colleagues on the same an identical or similar level within the company hierarchy

A single review object can be reviewed using more than one type of review

MANAGEMENT REVIEW

Review object is the entire project; checking whether project is on schedule and how management is doing

Often conducted when project reaches a particular planning milestone (i.e., a development phase is completed)

In agile projects, it takes the form of *retrospective meeting* where it is held following every sprint

INFORMAL REVIEW

Kind of soft review that nevertheless follows the standard review process without a strict formal structure

Minor issues can be resolved using this type of review

Usually initiated by the author

Success of informal review is highly dependent on the skills and motivation of reviewer

Different techniques can be used in informal review:

- *Buddy testing* – A teammate tries out your work
- *Pair programming* – Two people code together and review in real-time
- *Code swapping* – Developers swap code to review each other's work

WALKTHROUGH

Used to check for conformity with required standards and project specifications

A forum for discussing alternative implementations

Result of walkthrough should be a consensus opinion

Focus of a walkthrough is a meeting that is usually led by the author; results need to be recorded but there is no need to prepare formal transcript or summary report

Use of checklists is also optional

Little individual preparation is required as compared to technical review or inspection

CONT..

Reviewers use the comments and questions raised by the participants as a basis for identifying defects

This technique is useful for small teams (up to five people) and little or no preparation and follow-up is required

Can be used for checking non-critical objects

TECHNICAL REVIEW

Main focus is forming a consensus, evaluating product quality and building confidence in the review object

New ideas and suggestions for alternative implementations are welcome in a technical review

Colleagues of the author who work in the same or closely related domain should participate as reviewers; experts from other fields can help prevent from becoming blind to their own habits

Result need to be recorded, but not by the author

If discussion is not reaching any consensus, voting can be used and log the results in the review report

INSPECTION

Follow a strict predefined flow having entry and exit criteria

Here too, reaching a consensus is an important part of the process

Findings of the inspection should help the author to avoid similar mistakes in the future

Additional objective is the improvement of software development process

Fulfillment of entry criteria is verified before the inspection begins

CONT..

Reviewer's individual preparation takes place according to predefined rules

Some of the data collected during an inspection can also be used to identify the causes of weakness in the development process and thus to improve its quality

It is often referred to as design, code, or software inspection based on the type of documents that are being inspected

If formal review criteria exist, all type of documents can be inspected

SELECTING TYPE OF REVIEW

Depends on the objectives you are pursuing, the required quality, and effort this will cost

You have to decide from project to project which type of review is best suited to the situation at hand

Different questions can help answering which type of review is suitable:

- Does the review requires experts from different disciplines?
- Is scheduling easy or difficult?
- How much specific knowledge of the review object do the reviewers need?
- How formal is the review object?
- How much management support do you have?

BENEFITS OF REVIEW

Reviews are an efficient tool for assuring the quality of the work products under investigation

Early identification and correction of defects through reviews is usually cheaper than resolving defects later when application has reached an executable stage

Static tests are generally cheaper than dynamic tests

Reviews often saves significant amount of development time

Effort involved in making correction depends on the nature of the issue that requires attention

Dynamic test can involve less effort if a code review has already removed faults in the test object

Reviews can reveal inaccuracies in customer requirements that can be resolved in advance of implementation

As reviews takes place in group environment, they encourage exchange of knowledge among the participants

PEOPLE RELATED SUCCESS FACTORS

Select the *right* participants to reach your planned review objectives

It is useful to involve testers in the review, so they become familiar with project documents from the start of the development process and begin specifying right test cases

Participants should be given enough time to concentrate on details

If necessary, offer participants additional training in advance of the review especially for a formal review such as an inspection

If a review fails because of insufficient preparation, this is usually due to poor scheduling

If the reviewers don't understand the significance of the review or the effectiveness of the review process in quality improvement, you can use figures from the current project (or from earlier projects) to underscore the importance of the review process

DIFFERENCE BETWEEN STATIC AND DYNAMIC TESTING

Static tests identify faults directly in documents and other work products while dynamic tests usually identify failures in the source code in executable form

Dynamic testing verifies the visible, external behavior of the test object, whereas static tests focus on improving the internal quality of the object under investigation

TYPES OF DEFECTS

Requirement defects – Inconsistencies, ambiguities, contradictions, gaps, inaccuracies, or redundancies in requirements

Software design defects – Defects found in architecture documents that specifies the system's internal interfaces. Coupling and cohesion, algorithm and database structures are ineffectively designed

Coding defects – Defects occur due to inexperienced programmers. Detect redundant variables never invoked, unreachable duplicate code. It can be detected by a compiler or by using an automated static analysis tool

Deviations from standards – Insufficient adherence to programming guidelines leads to the poor quality; stick to them from start is crucial.

TYPES OF DEFECTS CONT..

Faulty interface specification - The names of the interfaces between components and their parameters (number, data type, and sequence) must be checked individually, as not all discrepancies will be found during integration

Security vulnerabilities – Static tests can also uncover these kinds of vulnerabilities like buffer overflow because constraints weren't explicitly checked or when input data or SQL queries are deliberately manipulated

Gaps or inaccuracies in test basis traceability or coverage – Connection between requirement and test case must be checked. A review report can point out missing tests for fulfilled exit criteria

CONCLUSION

Every work product (document) can be reviewed, provided that all the participants in the review are aware of how to read and understand the document in question

Reviews are static tests – can be applied at any stage

Documents are inspected by more than one person, and the results are discussed and recorded in a dedicated meeting

The activities involved in the review process are planning, initiation, individual preparation (i.e., individual review), discussion of the findings (issue communication and analysis, usually in a dedicated meeting), fixing and reporting

In order to better identify defects at the individual review stage, there are different review techniques you can use

CONCLUSION

The roles that need to be assigned for a review are manager, review leader, facilitator (moderator), author, reviewer (expert), and scribe

An informal review doesn't follow any formal rules, and the form of the resulting documents is not regulated. Because it involves relatively little effort, the informal review process has become a widely accepted and well-used technique

A walkthrough is an informal technique that sees the author presenting a document to reviewers in a meeting. Here too, preparation is minimal. Walkthroughs are ideal for small development teams who need to discuss alternatives and distribute know-how within the team

A technical review follows a more strictly regulated flow in which individual reviewer preparation is essential. Reviewers should be professional colleagues of the author so that alternative realizations can be discussed

CONCLUSION

An *inspection* is the most formal of these types of review. Preparation is based on checklists, and each review step has its own entry and exit criteria. The review session is led by a trained facilitator. Alongside checking the quality of the work product, the objective of an inspection is to improve the development and review processes

If you keep appropriate organizational and people-related success factors in mind, and as long as you take care to avoid known pitfalls, a review can be a seriously effective tool for increasing quality in software development projects

Reviews typically identify different defects than dynamic testing